Project Work

Important instructions related to project work–

1. Projects can be done in small groups of students.
2. It is compulsory for each student to do a minimum of 3 projects, one each in physics, chemistry and biology.
3. Project writing should follow a sequence. You can include pictures, paper cuttings, graphs, collections, photographs and other exhibits to elaborate.
4. During practical examinations, it is compulsory to conduct oral assessment of project work and experiments.
5. Project work may also be done on local issues.

Biology

1. Understand the process of classification.
2. Understand the different ways of treatment/testing/identifying symptoms of diseases.
3. Study the habitat of a plant.
4. Identify biodegradable and non-biodegradable wastes.

Chemistry

1. Collecting and making a list of different mixtures used in daily life and categorization into solutions, colloids and suspensions.
2. Making a list of elements, compounds and mixtures found around you and writing two uses of each one of them.
3. Understand the role of plastic codes in the recycling of objects made of plastics.
4. Ask 5 persons involved in different occupations in the locality of your school, whether their consumption of fossil fuels (coal, L.P.G. petrol, kerosene) has increased or decreased over time. Discuss in detail. Also find out the steps taken by them to reduce fossil fuel consumption.

Physics

1. Making a comparative study of examples from daily life related to acceleration, velocity and retardation.
2. Calculating the amount of electrical energy consumed daily at home/school.

Biology

Project work (Biodiversity and classification)

Objective : Understand the process of Classification

Materials required : Plastic scale, wooden scale, pencil, wooden block, alpin, rubber ball, cricket bat, key, rope, chalk, book, rubber, cycle tube, a piece of glass, plane mirror

Process: Write down the names of all these objects on the blackboard. Form two groups to start playing this game. The number of members of a team should not exceed 15. Choose a leader for each team. As you start playing, the leader of one of the teams must discuss with the team members and write the name of an object from the list on a piece of paper and hand it over to the teacher. The members of the other team should not get any idea of the chosen object. They have to find out the name of the object by asking appropriate questions the answer to which should be in 'yes' or 'no' only like is the object used for playing games? The questions must be asked by proper discussion within group members. Both groups should get a turn each in deciding the name of object. The group that asks a minimum number of questions to find out the object wins.

Question :

1. What were the characters that were made basis for asking the questions?
2. Make a flow chart of classification of the objects by taking the help of Figure1 of the chapter(Biodiversity and Classification).

Project Work (Our Health)

Objective : to conduct a survey to find out the number of people whogofor confirmation of disease before treatment (raise awareness towards need of tests and proper line of treatment).

- Contact any five people who are sick. Collect information from them on the basis of the following table and answer the questions given in this section.

Name of the sick person	Symptoms of the disease	Did the patient undergo test for confirmation? (Yes/No)	Name of the disease	Treatment

- How many patients resorted to confirmatory tests for the disease?
- How many patients underwent treatment?
- What were the different ways adopted for treatment?

Project Work (Habitat)

Objective : To study the habitat of a plant

Materials required : copy, pencil, rubber, sharpener, rope, hand lens, matchbox, bottle, paper bag etc.

Method : Divide into groups of 3-5 members. Let half of the group study Parthenium sp.(gajarghas)/grass and the other half study railway creeper(besharam)/Dhatura. Include the following points essentially to collect information-

- What were the plants found around an area of one square meter of the plant under study? Note their number. Animals feeding on these plants, try to collect some evidences in the form of the insect itself(be careful not to hold the insect rather collect the whole leaf on which the insect may be found to feed in a paper bag) or a leaf eaten by insect. Note the names of animals if you know, take the help of your teacher as well.
- Describe the kind of soil that you find in which the plant grows(like dry/wet, redsoil/blacksoil/etc.,sandy/clayey/etc.)
- Use of the plant if any.
- Collect any other information you find notable and worth observing.
- Try to make a sketch of the plant that was studied.

Get back to the class and share the informations collected with your friends.

Questions :

1. What were the similarities and differences found while comparing your observations?
2. There can be some difference in the habitat of the same species. Discuss this aspect on the basis of your findings.

Project Work (Waste and its Management)

Objective : Identify biodegradable and non- biodegradable wastes

Materials required : 8-10 earthen pots, soil, plain paper, gum, water, natural and human generated waste like vegetable or fruit peels, toffee papers, nails, glass pieces(be careful while handling them) or glass balls, covers of bottles etc.

Method : Follow the following method to find out about biodegradable and non-biodegradable wastes.

Fill soil upto a certain level in all the earthen pots. Add water to retain moisture. Separate out the waste materials and add them to the soil in the pots like add fruit and vegetable peels in one, a bottle cover in another, a nail in the third one etc. now cover up the materials with soil and add water over the same. Label the pots by writing the date and thetype of waste in it. Keep the pots in a shaded area for 3 to 4 weeks and leave them aside(remember to sprinkle water on them to keep the soil moist).

Dig out the waste material of all the pots after four weeks, observe the condition of the waste materials and write a report of your observation(use rubber gloves to carry out your work with soil). You could take the help of the following table for this.

S.No. of Pot	Name of object(waste)	Fully degraded	Partially degraded	No change
1.				
2.				
3.				
4.				

- What was the nature of objects that were fully degraded?
- What kind of waste is it?

Chemistry

Project work (Matter : Nature and Behaviour)

Objective : Classify the mixtures used in daily life into solutions, colloids or suspensions.

Theory : Solution is a homogenous mixture. The particles of a solution are so small that they cannot be seen through naked eye and we cannot seperate the particles of a solute from a solution by the process of filtration. As the particles are very small they do not disperse light rays. Therefore, we cannot observe the path of light rays in a solution.

Suspension is a heterogeneous mixture. We can see its particles through naked eye. These disperse light rays and hence we can see the path of light rays. We can seperate its particles through the process of filtration.

A colloid is a heterogenous mixture. Its particles are so small that we cannot see then by naked eyes. But they are large enough to disperse light rays. We cannot seperate its particles through filteration.

Procedure : Make a list of all the mixture that you use in your daily life.

S.No.	Substance	Solution/Colloid/Suspension	Basis of classification
1.	Lemonade	Solution	Cannot see the particles and cannot seperate them by filtration.
2.			
3.			
4.			
5.			
6.			
7.			
8.			
9.			
10.			

Discuss your observation in the classroom and on this basis write down point wise conclusions in your project report.

Note : Take examples of solution/colloids/suspension from your surroundings and items such as medicines etc.

While writing down the project report, include the difficulties faced by you and their solutions. If you had to make any changes in the procedure, do mention it. Also mention the things you learnt during this project.

Project work (Matter : Nature and Behaviour)

Objective : Classify the items in your surroundings into elements/compounds and mixture and write down 2 uses of each.

Theory : An element is a fundamental form of matter and cannot be divided into other simpler substances through chemical reactions. Elements can be classified as metals and non-metals. Metals have lustre, are good conductors of heat and electricity, ductile, malleable and produce metallic sounds. Non-metals do not show these properties. Non-metals are of different colour, are bad conductors of electricity and are brittle. Compounds are made from chemical combination

of 2 or more elements in a specific ratio. Properties of a compound are different from its constituent elements. Mixtures are made from two or more elements or compounds mixed in any ratio. Chemical reactions do not take place during the formation of mixtures. They possess the properties of constituent elements.

Procedure : Form your surroundings, classify items into elements/compounds/mixtures based on their properties and note your observations in the following table–

S.No.	Example	Element/ Compound/ Mixture	Constituents	Basis of classification	Uses
1.	Iron nail	Element	Iron	Cannot be divided into simpler susbstances.	
2.	Water	Compound	Hydrogen & Oxygen	Formed by chemical combination	
3.	Brass	Mixture	Copper and Zinc	Chemical reaction does not take place during formation.	
4.					
5.					
6.					
7.					
8.					
9.					

Project work (Coal, Petroleum and Petrochemicals)

Objective : To understand the role of plastic codes in the recycling of plastic items used in daily life.

Theory : In recent times, the plastic industry has rapidly grown as has the petrochemicals industry.We use a vast number of plastic objects in our daily life. On the basis of its property, plastic is coded and the code is mentioned on the item. We can easily see the code on plastic objects. Recycling of plastic is done on the basis of these codes.

Procedure :

- Formation of groups of students should take place under the supervision of teachers.
- Each group should visit at least 5 houses in their surrounding and gather information about the plastic items used by them and their plastic codes. Note down the plastic codes in your notebook. Discuss about the recycling of plastic.
- Students can do this survey in their locality.
- After the survey, discuss your observations and form a table of all the items under plastic codes 1 to 7 respectively.
- Make a report on the recycling process of all these tabulated items.
- Talk with the person of your neerby junkyard and collect information about the importance of plastic codes for them.
- The growing use of plastics is hazardous for our health. With the help of your head teacher arrange a discussion on this topic and include the points in your report.

Project work (Coal, Petroleum and Petrochemicals)

Objective : To know about the work done for conservation and use of fossil fuels. To encourage ourselves and society for the conservation of fossil fuels.

Theory : Fossil fuels are a source of energy which takes millions of years to form but our current stores will last only another 100 years. Therefore, we should try to stop misuse of fossil fuels. It will be available for future generations only if we use it wisely. We may face energy crisis in future therefore it is necessary to motivate society and community to use fossil fuels wisely.

Procedure :

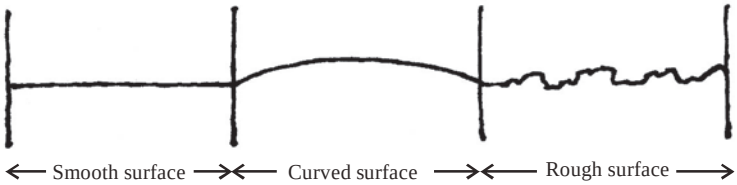
- Formation of groups should be done by the teacher.
- Each group must visit 5 houses near there school and have a discussion with 5 persons occuped in 5 different occupations. Gather information about their fossil fuel (coal, LPG, petrol, Kerosene) utilization. Ask whether their consumption has increased or decreased in last 5 years. Ask whether they take any steps for the conservation of fossil fuels and write down about it after a through discussion. Suggest them your own ideas about fossil fuel conservation. Note down your discussion in your project report.

Project Work (Physics)

Motion

Objective : To study velocity, acceleration and deceleration using daily life examples.

Materials required : Stop watch, different types of vehicles (cycles, motor-bike, car, auto), paper, pen, pencil, tape etc.



Theory : Distance covered in a given direction in a unit time by an object in motor is called velocity.

velocity = displacement / time

Acceleration : Rate of change of velocity in a non-uniform motion is called acceleration.

acceleration = change in velocity/time interval

Procedure :

1. Form a group of 4-5 members to study daily life examples of velocity, acceleration and deceleration.
2. Choose a path which has some part as plane surface, some part as curved surface and some rough patches.
3. Mark a point at some fixed distance. Ask one of the members to stand at that point.
4. Note down the distance and time of travel taken by various vehicles and fill in the table. Measure the distance using metre scale or tape and measure the time using a stopwatch.

Observe the figures carefully and study them.

S.No.	Description	Walking	Cycle	Auto	Bike	Car
1.	Time taken to cover 20 m. of plane surface.					
2.	Time taken to cover 20 m. of ..... surface.					
3.	Time taken to cover 20 m. of rough surface.					

Result : By doing a comparative study of the velocity, acceleration and deceleration of various vehicles we conclude that acceleration takes place on plane surface while deceleration takes place on rough surface.

Safety measure :

1. Use a common scale while marking distances so that we find regular distances and error free results.
2. Stand at a safer place on the roadside. to calculate the time. Be alert to avoid any mishappenings.

On the same basis you can take another example of walking on plane surface, climbing on stairs and climbing down from stairs.

Project work (work and energy)

Objective : Calculate the energy spent daily in your house/school.

Materials required : Watch, various electrical equipments, multi-meter etc.

Theory : We calculate the daily expenditure of energy using the following formula.

Total energy expended daily = No. of equipments × power of equipments (watts) × time (in hours) / 1000

Procedure : Calculate the energy spent daily at your house or school using the above formula.

Observation table

S.No.	Name of equipment	No. of equipment	Power of equipment	Time spent when the equipment was on	Energy spent	Total energy spent

Calculation :

1. Daily energy =
2. Monthly energy = daily energy × no. of days
3. Energy expenditure = total energy spent in month / rate.

Result : Energy spent daily in your house/school is .....

Safety measures :

1. Take proper safety measures for electricity use.
2. Don't touch any electrical equipment when it is on.
3. Stay away from equipments when they are in running condition.

Future scope : What measures can you take to avoid electrical energy wastage?